AD 2. AERODROMES

VABP AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VABP - RAJA BHOJ AIRPORT, BHOPAL

VABP AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	Aerodrome reference point coordinates and its site	231713N 0772013E 310.85 DEG / 654 M From Physical beginning of RWY 30	
2	Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves	288 DEG, 8KM from Bhopal Railway Station.	
3	Aerodrome elevation and reference temperature	1721 FT / 41.9 DEG C	
4	Magnetic variation, date of information and annual change	0.37 DEG W (2010) /0.033 DEG E	
5	Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available)	Airport Director, Airports Authority of India, RAJA BHOJ AIRPORT, Bhopal Airport, Bhopal -462030.,	
		Telephone:	(91) 755 2646001 (91) 9425008255
		Fax:	(91) 755 2646002 (91) 755 2646003 (91) 755 2640989
		AFS:	VABPYHYX
		Email:	APDBHOPAL@AAI.AERO
6	Types of traffic permitted (IFR/VFR)	IFR/VFR	
7	Remarks	Elevation in EGM08	

VABP AD 2.3 OPERATIONAL HOURS

1	Aerodrome Operator	MON-FRI 0400-1230 UTC (0930-1800 IST) SAT, SUN & HOL : NIL
2	Custom and immigration	Available on Prior Request
3	Health and sanitation	MI Room Available
4	AIS briefing office	H24
5	ATS reporting office (ARO)	As ATS
6	MET Briefing office	As ATS
7	Air Traffic Service	H24
8	Fuelling	As ATS
9	Handling	As ATS
10	Security	As ATS
11	De-icing	N/A
12	Remarks	1. The ATS approved hourly runway traffic handling capacity is as follows: Maximum number of arrival and departure -12 (the minimum spacing between two successive arrival and departure shall be more than five minutes) Maximum number of arrivals only -06 Maximum number of departures only -10

VABP AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	NIL
2	Fuel and Oil types	ATF (K-50),

3	Fuelling facilities and capacity	Indian Oil: Three Bowser (Capacity 9KL & 10KL,11KL), One Trolley Unit for AVGAS, 1KL Bharat Petroleum :Three Bowser (Capacity- 12KL,15KL,27KL)
4	De-icing facilities	NIL
5	Hangar space for visiting aircraft	NIL
6	Repair facilities for visiting aircraft	NIL
7	Remarks	NIL

VABP AD 2.5 PASSENGER FACILITIES

1	Hotel(s) at or in the vicinity of aerodrome	In the city.
2	Restaurant(s) at or in the vicinity of aerodrome	At AD and in the city
3	Transportation possibilities	Car Rental & Taxi
4	Medical Facilities	First aid at AD. Hospital in the city.
5	Bank and post office at or in the vicinity of aerodrome	Banks: Limited Hrs. Post office: Limited Hrs.
6	Tourist office	As ATS
7	Remarks	NIL

VABP AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	Aerodrome category for fire fighting	Within ATS HR: CAT-7
2	Rescue equipment	Available as per category.
3	Capability for removal of disabled aircraft	1. The critical aircraft for the airport is : A321 2. Primary responsibility for removal of disabled aircraft rests with the concerned airline. 3. The airlines may contact 'Aerodrome Coordinator for Disabled Aircraft Removal Operations' for coordinating with the agencies having Disabled Aircraft Recovery Kit. 4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations: Detail of Aerodrome coordinator : AIRPORT DIRECTOR Contact Nos. : +91-755-2646001, +91-9425008255 Email : apdbhopal@aai.aeroAs per Disabled Aircraft Removal Plan.
4	Remarks	NIL

VABP AD 2.7 SEASONAL AVAILABILITY CLEARING

1	Type(s) of clearing equipment	NIL
2	Clearance priorities	NIL
3	Remarks	NIL

VABP AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

1	Designation, surface and strength of aprons	Designator: New Apron Surface: Concrete Strength: PCN 80/R/C/W/T Designator: Old Apron Surface: Concrete Strength: PCN 60/R/B/W/T

2	Designation, width, surface and strength of taxiways	Designator: TWY A Width: 23 M Surface: Asphalt Strength: PCN 78/F/C/W/T Designator: TWY B Width: 23 M Surface: Concrete Strength: PCN 71/R/B/W/T Designator: TWY C Width: 23 M Surface: Asphalt Strength: PCN 15/F/C/W/T Designator: TWY D Width: 23 M Surface: Concrete Strength: PCN 69/R/B/W/T Designator: TWY E Width: 38 M Surface: Asphalt Strength: PCN 78/F/C/W/T Designator: TWY F Width: 23 M Surface: Asphalt Strength: PCN 80/F/C/W/T Designator: TWY G Width: 23 M Surface: Concrete Strength: PCN 80/R/C/W/T
3	Location and elevation of altimeter checkpoints	Location At apron Elevation 1711FT
4	Location of VOR checkpoints	On TWY E Coordinates: 231708.91940N 0772018.31909E, ELEV: 1720 FT
5	Position of INS checkpoints	NIL
6	Remarks	1. Isolation bay: Location: Southwest Length: 120M, Width: 110M, Shoulder: 7.5M all around. PCN 80/R/C/W/T Suitability: B767-400 2. Coordinates of Taxi Holding Position i. TWY D: Coordinates: 231711.50972N 0772018.81181E / ELEV: 1716FT ii. TWY E: Coordinates: 231708.84693N 0772018.20676E / ELEV: 1720FT iii. TWY F: Coordinates: 231704.90451N 0772012.06107E / ELEV: 1721FT iv. TWY G: Coordinates: 231719.69527N 0771955.06656E / ELEV: 1714FT 3. Coordinates of RWY & Taxi Intersection i. TWY G: Coordinates: 231717.05992N 0771953.51664E / ELEV: 1716FT ii. TWY E & TWY F: Coordinates: 2317086.71338N 0772014.88082E/ ELEV:1721FT 4. All elevations in EGM08

VABP AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands	Taxiing guidance provided on R/T. Guidelines at Apron. Nose-in Guidance at aircraft stands, A-VDGS provided on Parking stands 4, 5, 6, 7 and with aerobridge facility.
2	Runway and taxiway markings and lights	RWY Markings: THR ,Centreline, RWY Designation, Aiming point, RWY Edge Lights: Edge, End, THR, Centerline, TDZ RWY 30, Turn pad Edge Lights TWY Marking: Centerline, TWY Edge Marking, Holding positions at all TWY and RWY intersections. Lights: Edge.
3	Stop bars (if any)	TWY E, TWY F and TWY G
4	Remarks	

VABP AD 2.10 AERODROME OBSTACLES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
30/TKOF 12/APCH	TREE	231744.3N 0771841.0E	1777 FT	NO	Group Of Trees
30/APCH 12/TKOF	OTHER	231700.2N 0772033.7E	1717 FT	NO	Mobile Road Traffic (Road Elev 518.5 + 5 M Mobile Traffic)
30/APCH 12/TKOF	POLE	231652.7N 0772033.8E	1720 FT	NO	Light Post
30/APCH 12/TKOF	TREE	231701.2N 0772036.0E	1730 FT	NO	Group Of Trees
30/APCH 12/TKOF	TREE	231648.8N 0772038.3E	1745 FT	NO	Tree
30/APCH 12/TKOF	TREE	231652.2N 0772038.5E	1731 FT	NO	Tree
30/APCH 12/TKOF	POLE	231700.0N 0772040.3E	1731 FT	NO	Electric Pole
30/APCH 12/TKOF	TREE	231658.6N 0772040.9E	1733 FT	NO	Tree
30/APCH 12/TKOF	TREE	231659.5N 0772042.6E	1752 FT	NO	Group Of Trees
30/APCH 12/TKOF	TREE	231650.0N 0772042.9E	1747 FT	NO	Group Of Trees
30/APCH 12/TKOF	TREE	231658.4N 0772044.0E	1750 FT	NO	Group Of Trees
30/APCH 12/TKOF	TREE	231654.3N 0772048.6E	1750 FT	NO	Group Of Trees
30/APCH 12/TKOF	TREE	231652.0N 0772051.2E	1762 FT	NO	Group Of Trees
30/APCH 12/TKOF	BUILDING	231647.5N 0772047.3E	1762 FT	NO	Building
30/APCH 12/TKOF	OTHER	231640.5N 0772111.9E	1801 FT	NO	Mast

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
30/APCH 12/TKOF	TREE	231652.6N 0772052.6E	1755 FT	NO	Group Of Trees
In circling area and at AD	OTHER	231659.4N 0772020.8E	1734 FT	LGTD	GP Monitor Antenna
In circling area and at AD	TREE	231705.1N 0772028.3E	1759 FT	NO	Group Of Trees
In circling area and at AD	BUILDING	231705.0N 0772028.6E	1740 FT	NO	Building
In circling area and at AD	FENCE	231701.8N 0772032.2E	1717 FT	NO	Airport Boundary Wall With Fencing On Top
In circling area and at AD	POLE	231701.0N 0772034.0E	1714 FT	NO	Light Post
In circling area and at AD	POLE	231704.3N 0772029.7E	1742 FT	NO	Light Pole
In circling area and at AD	POLE	231703.8N 0772031.1E	1739 FT	NO	Light Pole
In circling area and at AD	TREE	231724.3N 0771921.6E	1800 FT	NO	Group Of Trees
In circling area and at AD	POLE	231728.3N 0771917.5E	1738 FT	NO	Light Pole
In circling area and at AD	TREE	231727.8N 0771917.1E	1781 FT	NO	Group Of Trees
In circling area and at AD	TREE	231726.3N 0771916.9E	1796 FT	NO	Group Of Trees
In circling area and at AD	BUILDING	231728.0N 0771915.8E	1764 FT	NO	Temple
In circling area and at AD	POLE	231729.1N 0771915.8E	1737 FT	NO	Light Pole
In circling area and at AD	TREE	231728.6N 0771913.0E	1812 FT	NO	Group Of Trees
In circling area and at AD	TREE	231730.4N 0771909.2E	1786 FT	NO	Group Of Trees
In circling area and at AD	TREE	231732.7N 0771905.1E	1782 FT	NO	Group Of Trees
In circling area and at AD	POLE	231734.0N 0771903.3E	1744 FT	NO	Electric Pole
In circling area and at AD	POLE	231734.8N 0771901.5E	1746 FT	NO	Electric Pole
In circling area and at AD	TREE	231733.6N 0771900.6E	1801 FT	NO	Group Of Trees
In circling area and at AD	POLE	231735.6N 0771859.7E	1749 FT	NO	Electric Pole
In circling area and at AD	TREE	231736.6N 0771853.9E	1786 FT	NO	Group Of Trees
In circling area and at AD	TREE	231740.3N 0771847.3E	1761 FT	NO	Group Of Trees
In circling area and at AD	TREE	231741.3N 0771846.6E	1767 FT	NO	Group Of Trees
In circling area and at AD	OTHER	231706.4N 0772027.6E	1763 FT	NO	Flood Light
In circling area and at AD	BUILDING	231706.5N 0772028.1E	1739 FT	NO	Old Terminal Building

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	ANTENNA	231706.9N 0772029.6E	1754 FT	NO	Antenna On Old Terminal Building
In circling area and at AD	OTHER	231707.7N 0772029.7E	1768 FT	NO	Flood Light
In circling area and at AD	BUILDING	231707.4N 0772029.8E	1773 FT	NO	ATC Top
In circling area and at AD	OTHER	231707.3N 0772029.8E	1794 FT	NO	Lightning Arrester On ATC Building
In circling area and at AD	OTHER	231707.5N 0772029.9E	1777 FT	NO	Aerodrome Beacon (ATC Tower)
In circling area and at AD	ANTENNA	231706.6N 0772030.1E	1747 FT	NO	Dish Antenna On Old Terminal Building
In circling area and at AD	ANTENNA	231706.8N 0772030.4E	1749 FT	NO	Dish Antenna On Old Terminal Building
In circling area and at AD	OTHER	231704.8N 0772030.5E	1739 FT	NO	Flood Light
In circling area and at AD	OTHER	231705.9N 0772032.2E	1771 FT	NO	Flood Light
In circling area and at AD	POLE	231652.6N 0772032.3E	1719 FT	NO	Light Post
In circling area and at AD	POLE	231704.7N 0772032.9E	1736 FT	NO	Light Post
In circling area and at AD	BUILDING	231704.3N 0772035.3E	1755 FT	NO	Met Office
In circling area and at AD	POLE	231701.1N 0772038.7E	1722 FT	NO	Electric Pole
In circling area and at AD	BUILDING	231644.7N 0772042.3E	1762 FT	NO	Building
In circling area and at AD	TREE	231700.2N 0772045.3E	1772 FT	NO	Mango Tree
In circling area and at AD	POLE	231704.3N 0772032.0E	1739 FT	NO	Light Pole
In circling area and at AD	POLE	231705.4N 0772031.5E	1742 FT	NO	Light Pole
In circling area and at AD	ANTENNA	231706.6N 0772030.7E	1762 FT	NO	Ant on Bldg (Building Top = 530.8 M)
In circling area and at AD	BUILDING	231657.6N 0772051.5E	1759 FT	NO	Building U/C
In circling area and at AD	BUILDING	231624.3N 0772241.5E	1988 FT	NO	Building
In circling area and at AD	OTHER	231622.3N 0772242.8E	1985 FT	NO	Mast
In circling area and at AD	BUILDING	231650.0N 0772150.9E	1890 FT	NO	Temple
In circling area and at AD	OTHER	231627.2N 0772244.3E	1949 FT	NO	Mast
In circling area and at AD	OTHER	231635.4N 0772237.2E	1962 FT	NO	Mast

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	231637.5N 0772239.7E	1939 FT	NO	Mast
In circling area and at AD	BUILDING	231623.9N 0772250.4E	1939 FT	NO	Nehru Hospital
In circling area and at AD	BUILDING	231620.8N 0772301.6E	1946 FT	NO	School Building
In circling area and at AD	OTHER	231621.2N 0772246.1E	1998 FT	NO	Mast
In circling area and at AD	BUILDING	231622.9N 0772258.3E	1949 FT	NO	Building
In circling area and at AD	OTHER	231624.8N 0772301.0E	1991 FT	NO	Mast
In circling area and at AD	BUILDING	231633.5N 0772249.1E	1929 FT	NO	Building
In circling area and at AD	OTHER	231722.5N 0771933.6E	1730 FT	LGTD	W.D.I.
In circling area and at AD	OTHER	231659.4N 0772020.8E	1734 FT	LGTD	Glide Path Monitor Antenna Old
In circling area and at AD	OTHER	231700.4N 0772018.7E	1759 FT	NO	Glide Path Antenna Old
In circling area and at AD	BUILDING	231700.6N 0772018.3E	1726 FT	NO	Glide Path Hut Old
In circling area and at AD	OTHER	231715.6N 0772007.3E	1736 FT	LGTD	W.D.I.
In circling area and at AD	FENCE	231711.9N 0771954.6E	1721 FT	NO	Airport Boundary Wall Fencing On Top
In circling area and at AD	OTHER	231700.1N 0772019.3E	1768 FT	LGTD	Glide Path Main Antenna
In circling area and at AD	OTHER	231700.0N 0772019.3E	1733 FT	LGTD	DME Antenna (Co-Located With Glide Path)
In circling area and at AD	OTHER	231659.2N 0772021.1E	1731 FT	LGTD	Glide Path Monitor Antenna
In circling area and at AD	OTHER	231700.1N 0772019.1E	1729 FT	NO	Lightning Arrester On Glide Path Hut (Building Top = 526.14)
In circling area and at AD	ANTENNA	231705.6N 0772026.2E	1748 FT	LGTD	Met Antenna
In circling area and at AD	OTHER	231702.1N 0772021.6E	1719 FT	NO	PAPI Light
In circling area and at AD	SIGN	231704.3N 0772012.6E	1724 FT	NO	Sign Board
In circling area and at AD	SIGN	231704.9N 0772016.2E	1723 FT	NO	Sign Board
In circling area and at AD	SIGN	231719.4N 0771951.3E	1718 FT	NO	Sign Board
In circling area and at AD	SIGN	231720.0N 0771954.1E	1717 FT	NO	Sign Board
In circling area and at AD	SIGN	231720.1N 0771954.3E	1717 FT	NO	Sign Board

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	SIGN	231719.3N 0771956.0E	1717 FT	NO	Sign Board
In circling area and at AD	SIGN	231709.1N 0772012.6E	1724 FT	NO	Sign Board
In circling area and at AD	SIGN	231710.4N 0772013.3E	1722 FT	NO	Sign Board
In circling area and at AD	SIGN	231709.2N 0772016.6E	1722 FT	NO	Sign Board
In circling area and at AD	SIGN	231709.8N 0772017.4E	1721 FT	NO	Sign Board
In circling area and at AD	TREE	231722.7N 0771928.6E	1764 FT	NO	Tree
In circling area and at AD	TREE	231721.9N 0771930.5E	1800 FT	NO	Tree
In circling area and at AD	TREE	231720.7N 0771933.3E	1792 FT	NO	Tree
In circling area and at AD	TREE	231720.4N 0771934.1E	1811 FT	NO	Group Of Trees
In circling area and at AD	POLE	231720.1N 0771935.0E	1721 FT	NO	Light Pole
In circling area and at AD	TREE	231719.5N 0771935.6E	1789 FT	NO	Tree
In circling area and at AD	TREE	231719.1N 0771937.3E	1755 FT	NO	Tree
In circling area and at AD	POLE	231718.9N 0771937.9E	1720 FT	NO	Light Pole
In circling area and at AD	POLE	231718.4N 0771939.0E	1720 FT	NO	Light Pole
In circling area and at AD	POLE	231717.9N 0771940.2E	1721 FT	NO	Light Pole
In circling area and at AD	OTHER	231711.5N 0771947.1E	1777 FT	NO	Over Head Water Tank
In circling area and at AD	TREE	231726.0N 0771920.6E	1799 FT	NO	Group Of Trees
In circling area and at AD	TREE	231724.0N 0771922.7E	1821 FT	NO	Group Of Trees
In circling area and at AD	OTHER	231716.2N 0771942.9E	1740 FT	NO	Arch
In circling area and at AD	TREE	231716.0N 0771944.0E	1776 FT	NO	Group Of Trees
In circling area and at AD	TREE	231709.8N 0771956.6E	1743 FT	NO	Group Of Trees
In circling area and at AD	TREE	231706.2N 0771959.9E	1760 FT	NO	Group Of Trees
In circling area and at AD	OTHER	231617.8N 0772222.9E	1910 FT	NO	Pylon Mast
In circling area and at AD	BUILDING	231737.1N 0772146.9E	2006 FT	NO	Building On Hill
In circling area and at AD	BUILDING	231825.5N 0772150.3E	1886 FT	NO	Building On Hill
In circling area and at AD	BUILDING	231659.8N 0772150.5E	2118 FT	NO	Temple On Hill

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	231701.9N 0772151.0E	2128 FT	NO	Cellphone Mast
In circling area and at AD	POLE	231727.1N 0772153.9E	1972 FT	NO	Flag Pole
In circling area and at AD	OTHER	231717.3N 0772156.2E	1970 FT	NO	Over Head Water Tank
In circling area and at AD	OTHER	231635.3N 0772237.3E	1939 FT	NO	Cellphone Mast
In circling area and at AD	POLE	231656.0N 0772153.6E	2082 FT	NO	Rope Way Pole
In circling area and at AD	OTHER	231348.4N 0772311.1E	2215 FT	NO	Police Mast
In circling area and at AD	OTHER	231422.4N 0772317.9E	2257 FT	LGTD	TV Tower
In circling area and at AD	OTHER	231701.6N 0772021.3E	1719 FT	NO	PAPI Light
In circling area and at AD	OTHER	231701.8N 0772021.4E	1719 FT	NO	PAPI Light
In circling area and at AD	OTHER	231704.5N 0772035.2E	1755 FT	NO	Lightning Arrester on Building (Building Top = 530.8 M)

VABP AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Name of the associated meteorological office	Bhopal
2	Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours	H-24
3	Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts	Bhopal 9 AND 24HRS
4	Availability of the trend forecast for the aerodrome and interval of issuance	Trend 30 MIN
5	Information on how briefing and/or consultation is provided	Provided
6	Types of flight documentation supplied and language(s) used in flight documentation	Tabular Form (English)
7	Charts and other information displayed or available for briefing or consultation	S,U85,U70,U50,U30,U20
8	Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images;	RVR equipment and Digital Current Weather Instrument System (DCWIS) are available for RWY 12/30 providing RVR information for TDZ RWY 12 and TDZ RWY 30. Instrument RVR at MID RWY 12/30 Available
9	The air traffic services unit(s) provided with meteorological information	Bhopal ATC and ACS
10	Additional information, e.g. concerning any limitation of service.	NIL

VABP AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations	TRUE Bearings	Dimensions of RWY (M)	Strength of pavement (PCN) and associated data) and surface of runway and associated stopways	Geographical coordinates for threshold and runway end
1	2	3	4	5
12	117.66 DEG	2744 x 45 M	88/F/C/W/T Asphalt	THR: 231740.51N 0771905.08E RWY END: 231659.09N 0772030.62E
30	297.66 DEG	2744 x 45 M	88/F/C/W/T Asphalt	THR: 231659.09N 0772030.62E RWY END: 231740.51N 0771905.08E

THR elevation and highest elevation of TDZ of precision APP RWY	Slope of runway and associated stopway	Dimensions of stopway (M)	Dimensions of clearway (M)	Dimensions of strips (M)
6	7	8	9	10
THR: 1714.0FT TDZ: 1720.0FT	-0.02%	NIL		2864 x 280 M
THR: 1712.0FT TDZ: 1721.0FT	0.02%	NIL		2864 x 280 M

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
240M x 90M		NIL	i. PCN RWY 12: 0-699M: 63/F/B/W/T 699-2744M: 88/F/C/W/T ii. Between TWY E and TWY G the strip varies from 135 to 140M iii. Elevation in EGM08
90M x 90M		NIL	i. PCN RWY 30: 0-2045M: 88/F/C/W/T 2045M-2744M: 63/F/B/W/T ii. Elevation in EGM08

VABP AD 2.13 DECLARED DISTANCES

RWY Designator	Take-off run available TORA (M)	Take-off distance available TODA (M)	Accelerate distance available ASDA (M)	Landing distance available LDA (M)	Remarks (including runway entry or start point where alternative reduced declared distances have been declared)
1	2	3	4	5	6
12	2744	2744	2744	2744	---
30	2744	2744	2744	2744	

VABP AD 2.14 APPROACH AND RUNWAY LIGHTING

Runway Designator	Type, length and intensity of approach lighting system	Runway threshold lights, colour and wing bars	Type of visual slope indicator system	Length of runway touchdown zone lights
1	2	3	4	5
12	SALS 420 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (52.97FT)	
30	CAT II 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (55.61FT)	900 M

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
2744 M 15 M LIH White	2744 M 60 M White LIH	Red		Cross bar consisting of 11 LGT fittings at a distance of 300M from THR RWY 12.
2744 M 15 M LIH White	2744 M 60 M White LIH	Red		1. CAT II APCH Lights with Cross bar, White colour provided at distance of 150M, 300M, 450M, 600M and 750M from THR RWY 30. Side rows of red colour provided up to 270M from THR RWY 30. 2. Colour of RWY Centreline lights White from Threshold to 900M from RWY end. Alternate variable White and Red from 900M to 300M from RWY end. Red from 300M to RWY end. 3. Colour of RWY Edge lights White from Threshold to 600M from RWY end. Yellow from 600M to RWY end. 4. TDZ Lights spacing 30M intensity High. 5.Cross bar provided at distance of 150M, 300M, 450M, 600M and 750M from THR RWY 30

VABP AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any)	ABN	At ATC Tower Building FLG White & Green FREQ 24-30 revolution per minute. HO
		IBN	NIL
2	Location and lighting (if any) of anemometer/ landing direction indicator;	LDI	Near signal area lighted
		Anemometer	On tower building.not lighted
3	Taxiway edge and taxiway centre line lights;	Edge	AVBL
		Centre Line	NIL
4	Secondary power supply including switch-over time;	Secondary power supply to all lighting at AD. Switch-over time (except Apron Flood Lights & Isolation Bay Flood lights): Less than 01 SEC. Switch-over time (Apron Flood Lights & Isolation Bay Flood lights): Less than 15 SEC.	
5	Remarks	RWY Guard LGT Configuration 'A' available at TWY E, TWY G and TWY F. Apron Flood Lights and Apron Edge Light (Blue Colour) available.	

VABP AD 2.16 HELICOPTER LANDING AREA

1	Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area	Not Established
2	TLOF and/or FATO area elevation:	Not Established
3	TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking;	Not Established
4	True bearings of FATO;	Not Established
5	Declared distances available	Not Established
6	Approach and FATO lighting;	Not Established
7	Remarks	Not Established

VABP AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

1	Airspace designation, geographical coordinates and lateral limits	CTR: Circular area centered on 231713N 0772013E within a 11NM radius.
2	Vertical limits	FL 75
3	Airspace classification	D
4	Call sign and language(s) of the air traffic services unit providing service;	Bhopal Tower, English
5	Transition altitude	5000 FT
6	Hours of applicability	H24
7	Remarks	Bhopal Approach Control Unit is combined with Bhopal Aerodrome Control Tower.

VABP AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available
1	**2**	**3**	**4**
APP	Bhopal Tower	118.050 MHZ	
APP	Bhopal Tower	118.550 MHZ	
TWR	Bhopal Tower	118.050 MHZ	
TWR	Bhopal Tower	118.550 MHZ	
ATIS	Bhopal Information	127.250 MHZ	
ALRS	Emergency Frequency	121.500 MHZ	NIL

Logon address, as appropriate	Hours of operation	Remarks
5	**6**	**7**
	H24	1. Main Frequency 2.Functions of Approach Control Service provided by Aerodrome Control Tower.
	H24	1. Backup Frequency 2. Functions of Approach Control Service provided by Aerodrome Control Tower.
	H24	NIL
	H24	ALT FREQ

Logon address, as appropriate	Hours of operation	Remarks
	H24	NIL
	H24	NIL

VABP AD 2.19 RADIO NAVIGATION AND LANDING AIDS

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
1	2	3	4
LOC 30 CAT II	IBPH	109.900 MHz	H24
GP 30	IBPH	333.800 MHz	H24
DME ILS 30	IBPH	CH36X	H24
DVOR/DME	BPL	117.100 MHz CH118X	H24

Geographical coordinates of the position of the transmitting antenna	Elevation of transmitting antenna of DME/ elevation of GBAS reference point	Service volume radius from the GBAS reference point	Remarks
5	6	7	8
231746.7N 0771852.3E			
231700.1N 0772019.3E			3 DEG
231700.1N 0772019.3E	1733 FT		Collocated with GP 30. Elevation in EGM08
231659.7N 0772011.9E	1749 FT		Elevation in EGM08

VABP AD 2.20 LOCAL AERODROME REGULATIONS

I. Apron Procedures for VVIP Flight (Type B777- 300ER):

i. Temporary Parking for VVIP B777-300 ER type Aircraft (Reduced Take-off Weight) is aircraft stand no. 01, 02 and 03 at New Apron (Facing southeast direction)

ii. Aircraft will be parked facing south.

iii. PCN Value of New Extended Turnpad-Suitable for VVIP B777-300 ER type Aircraft (at Reduced Take-off weight.):

RWY30 - 88/ F/C/W/T (Designed Value)

RWY12 - 88/ F/C/W/T (Designed Value)

II. Taxi Procedures VVIP B777-300 ER Type Aircraft (Reduced Take-off Weight):

i. Mode of Operation – Power in Power out

ii. Temporary Parking at aircraft stand 01,02 & 03 of New Apron (facing Southeast Direction)

iii. Lead in via TWY G to temporary parking stand of new apron.

iv. Lead out via TWY G to holding point.

v. Parking stand 01,02, 03 and 04 of New Apron will be kept vacant.

vi. Vehicular movement on service road adjacent to the Parking stand 01,02,03 and 04 of New Apron will not be allowed during taxi in and taxi out of the aircraft.

vii. Follow me service will be provided during Taxi in and Taxi out.

III. No Aircraft shall take lock turn/ pivotal turn on the Runway to avoid pavement damage.

VABP AD 2.21 NOISE ABATEMENT PROCEDURES

VABP AD 2.22 FLIGHT PROCEDURES

RADIO COMMUNICATION FAILURE PROCDURES

1. INTRODUCTION

Radio communication failure procedures are described in ICAO's PANS- ATM (DOC 4444), Manual of Air Traffic Services Part-I and DGCA CAR Section 9, Series C, Part 1.

2. GENERAL

2.1 Radio communication failure may occur due to:

a) Airborne Transmitter and/or Receiver Failure

b) Ground station Transmitter and/or Receiver Failure.

2.2 As soon as it is known to ATC that two-way communication has failed, the action shall be taken to ascertain whether the aircraft is able to receive transmissions from the air traffic control unit by requesting it to execute a specified manoeuvre or to transmit if possible, a specified signal in order to indicate acknowledgement.

2.3 All Transponder equipped aircraft experiencing Radio Communication Failure shall set transponder to Mode A/C code 7600 as soon as practicable

(Note: This requirement of setting transponder to Mode A/C code 7600 in no way imposes any restriction on the pilot's decision to set transponder to Mode A/C code 7500 or 7700, whenever required).

2.4 Immediately after the detection of RCF, the aircraft shall attempt to establish communications with the appropriate air traffic control unit using all other available means. (Bhopal Tower- Main and Standby/ Indore Tower (adjacent unit)/Nagpur control/Airline's channel/Emergency 121.5Mhz/ suitable HF frequency)

2.5 In addition, the aircraft shall keep watch for any instructions that may be issued by visual signals directed to aircraft from ground.

2.6 The pilot shall make blind transmission to ATC of all necessary reports and actions taken by the aircraft, e.g. descent, turn, joining hold, leaving for approach, FAF etc., irrespective of whether partial/complete RCF has been established or not.

2.7 Action shall be taken by Bhopal ATC to reach aircraft, which were under his control, via all available means, which may include Manpack VHF Set, relay through other aircrafts, through airlines' channel (company frequency or satphone, mobile/ cell phone, or pager on board), adjacent ATS unit etc. Action shall be taken to ascertain whether the aircraft is able to receive transmissions from the air traffic control unit by requesting it to execute specified manoeuvre which can be observed either visually (if the aircraft is visible to ATC) or Nagpur RADAR display (through Nagpur ACC). Further actions, as described hereunder, may be followed, as and when necessary.

2.8 If the aircraft can be observed from a control tower, ATC shall determine if the aircraft receiver is functioning by:

i) instructing the pilot to manoeuvre the aircraft in such a manner as to alternately raise and lower the wings in a rocking motion;

ii) if observed from a control tower by night, by instructing the pilot to flash the landing and navigation lights twice.

2.9 Aircraft experiencing radio communication failure shall be provided all required information necessary for the safe and efficient conduct of flight and landing clearance without expecting acknowledgment.

2.10 After landing at Bhopal Airport, Aircraft shall lookout for follow me jeep on nearest taxiway holding point and after sighting the Follow me Jeep, Aircraft shall follow the Follow Me Jeep for parking to the allotted bay. Upon sighting the Marshaller, aircraft shall stop following "Follow me Jeep" and shall follow marshaller for parking.

2.11 When Bhopal ATC becomes aware of the RCF of the aircraft, the controller concerned shall accord priority for the RCF aircraft to land.

2.12 If the aircraft re-establishes communication or has landed the ATC unit shall inform all concerned units/ aerodromes along the route as per the flight plan including the alternate aerodrome.

3. ASSIGNED RWY AND ITS AVAILABILITY FOR RCF AIRCRAFT

3.1 In case of an arriving aircraft experiencing RCF, when Runway for landing has already been advised to the aircraft by ATC, such runway shall be considered as the assigned runway. If an arriving aircraft has not been advised of any runway, Runway 30 shall be considered as assigned runway for such arrival.

3.2 In case of departures from Bhopal returning on account of RCF, the departure runway of such aircraft shall be considered as assigned runway for landing.

3.3 If Radio communication failure occurs after the final/interception turn, the aircraft may continue its approach to land on such Runway.

3.4 Runway lights and Approach lights in 'SWITCHED ON' position shall indicate the availability of such Runway for aircraft experiencing RCF.

3.5 In strong tail wind conditions for assigned Runway, aircraft experiencing RCF will carry out a missed approach on assigned runway and after following complete missed approach for such approach, aircraft will carry out either visual or Instrument approach procedure published for the opposite runway.

4. PROCEDURES FOR ARRIVALS

4.1 In case Radio Communication failure takes place after establishing final approach track, aircraft may continue the approach and land if visual or GO AROUND and carry out the missed approach procedure and join the VOR (117.1) BPL Holding procedure climbing to and maintaining 3800 feet.

4.2 Except when descent clearance has already been received from ATC and acknowledged, pilot shall continue on ATS route cleared to BPL maintaining filed flight plan level or level cleared by adjacent Area control unit (Nagpur ACC) and do not commence descent before 25 miles VOR (117.1) BPL.

4.3 After crossing 25 miles, the pilot shall descend to FL 080 to join the published hold over VOR (117.1) BPL for the runway as mentioned in para 3 above.

4.4 Commence descent to 3800 feet in published Hold over VOR (117.1) BPL at or as close as possible to, Expected Approach Time (EAT) last received and acknowledged; or if no Expected approach time (EAT) has been received and acknowledged, at or as close as possible to, Estimated time of arrival (ETA) resulting from the current flight plan. 4.5 Leave VOR (117.1) BPL and carry out published ILS/VOR procedure for runway as mentioned in para 3

4.5 Leave VOR (117.1) BPL and carry out published ILS/VOR procedure for runway as mentioned in para 3.

4.6 Land, if possible, within 30 minutes after the estimated time of arrival (ETA) or the last acknowledged expected approach Time (EAT), whichever is later.

5. PROCEDURES FOR DEPARTURE

5.1 DEPARTURE INTENDING TO CONTINUE TO DESTINATION

5.1.1 Any departing aircraft experiencing RCF and intending to continue to its filed Flight plan destination shall:

i) Continue ATS route as per filed flight plan, climbing to/maintaining cleared and acknowledged flight level, or minimum flight altitude if higher, for a period of 20 minutes following the aircraft's failure to report its position over compulsory reporting point and thereafter adjust level and speed in accordance with the filed flight plan to destination.

ii) When the aircraft is given clearance for offset from flight plan route without a specified limit, re-join the current flight plan route after passing 20 Mile from VOR (117.1) BPL but no later than the next significant point, taking into consideration the applicable minimum flight altitude.

5.2 DEPARTURE INTENDING TO LAND BACK AT BHOPAL AIRPORT

5.2.1 Any departing aircraft experiencing RCF after departure and intending to land back at Bhopal Airport, shall:

i) Maintain the last assigned level, or minimum flight altitude if higher, for a period of 10 minutes on the flight plan route or on offset radial as cleared in ATC clearance.

ii) After 10 minutes turn towards VOR (117.1) BPL via the same flight plan route or radial as per departure clearance. On Reaching VOR (117.1) BPL joins hold as published.

iii) If required, jettison fuel, then remain in hold over VOR (117.1) BPL after taking all necessary precautions.

iv) Commence descent in the hold to 3800 feet. Leave VOR (117.1) BPL and carry out the published ILS/VOR Procedure for the runway as mentioned in para 3 above.

6. PROCEDURES FOR VFR TRAFFIC

6.1 In case of a VFR flight experiencing RCF and does not intend to land at Bhopal airport:

i) It shall continue to fly in VMC, land at the nearest suitable aerodrome /helipad, and report its arrival by the most expeditious means to the Bhopal ATC.

6.2 In case of a VFR flight experiencing RCF and intending to land at Bhopal airport:

i) Proceed according to the current flight plan route in VMC and maintain the altitude/level last assigned and acknowledged.

ii) Proceed towards 15 miles South-West or North-East of the Bhopal airfield subject to the arrival route and descent to 1500 ft AGL.

iii) Maintaining 1500FT AGL proceed towards Bhopal airport to join the traffic circuit (at Downwind leg) for the last acknowledged Runway or the Runway as per para 3 above.

iv) Carry out a complete standard aerodrome traffic circuit and land on the last acknowledged Runway or the Runway as per para 3.

6. VARIOUS SIGNALS USED BY ATCO DURING COMMUNICATION FAILURE

The various signals used by the ATCOs during communication failure are as follows.	**Aircraft in Flight**	**Aircraft on the ground**
Steady green	Cleared to land	Cleared for take-off
Steady red	Give way to other aircraft, continue circling	Stop
A series of green flashes	Return for landing* (*Clearances to land and to taxi will be given in due course.)	Cleared to taxi
Series of red flashes	Aerodrome is unsafe, do not land	Taxi clear of landing area in use
Series of white flashes	Land at this aerodrome and proceed to the apron*	Return to starting point on the aerodrome
Red pyrotechnic	Notwithstanding any previous instructions, do not land for the time being	
Flashing runway or taxiway lights	Vacate the runway and observe the tower for light signal	

LOW VISIBILITY PROCEDURES
RAJA BHOJ AIRPORT, BHOPAL (VABP)
1 INTRODUCTION
1.1 As per the latest amendment of DGCA Civil Aviation Requirements (CAR) section 8, Series 'C', Part-I on All-Weather Operations, (Rev. 10, issue Oct 2022), Low visibility procedures are required at aerodromes for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs (LVTO).
1.2 Para 5.3 of DGCA CAR Section 8, Series C, Part-1 says, "An operator shall not conduct take-off with RVR/visibility less than standard Category I conditions of 550 m RVR/800 m visibility unless low visibility procedures are enforced".
1.3 Further, as per para 5.2 of DGCA CAR Section 8, Series C, Part-1 on All Weather Operations, "An operator shall not conduct low visibility take-offs in less than 400 m RVR unless approved by DGCA. Operators may be authorized LVTO minima of up to 75m. Foreign operators, who are authorized by their State Regulatory Authority for LVTO, shall submit requisite documents to DGCA for approval of LVTO at Indian aerodromes."
1.4 As per para 6.1 of DGCA CAR Section 8, Series C, Part-1 on All Weather Operations, "An operator shall not use an aerodrome for Category II or III operations unless the aerodrome is approved for such operations"; and further as per Para 6.2 "An operator shall verify that low visibility procedures (LVP) have been established, and will be enforced, at those aerodromes where low visibility operations are to be conducted."
1.5 These provisions necessitated the need for low visibility procedures for accommodating/permitting arrivals and departures in Visibility/RVR less than 800M/550M at Raja Bhoj Airport, Bhopal.
1.6 As per provisions of Aerodrome Design and Operations CAR and ICAO Annex 14,Runway centre line lights are required for take-off in RVR below 400 M. Further, as per Aerodrome Design and operations CAR para 9.8.7, Surface Movement Radar (SMR) need to be provided at an aerodrome intended for use in Runway Visual Range conditions less than a value of 350 m.
1.7 Accordingly, as per the provisions of DGCA CAR Section 8, Series C, Part-1 on All Weather Operations (Rev. 10, issue Oct 2022); DGCA CAR Section 8, Series B, Part 1 on Approval for ILS Cat-II/III Operations dated 14.06.2023, DGCA CAR Section 4, Series B, Part-1 on Aerodrome Design and Operations; ICAO DOC 9365; DGCA Aerodrome Advisory Circular AD AC NO. 4 of 2022; and DGCA letter no. AV1503/02/03-AL dated 12.01.2018; Low visibility Procedures have been developed for Bhopal Airport to accommodate/permit Arrivals and departures in Visibility/RVR less than 800 m/550 m up to 350 m RVR.
2 PURPOSE
2.1 The purpose of this document is to define the Low Visibility Procedures during CAT II Approaches and/or Low visibility Take-offs at Bhopal Airport.
3 SCOPE
3.1 This is applicable to all the concerned personnel involved in the Low Visibility Procedures as mentioned in this SOP.
3.2 The Low Visibility Procedure (LVP) describe the procedures and actions that are required to be taken by the Air Traffic Control (ATC), Pilots, Airline Operators, MET dept, Vehicle Operators, Airport Operations, Electrical & Civil Engg., CNS, Airport Fire Services and Ground Handling Agencies during Low Visibility operations at Bhopal Airport.
3.3 The procedures contained in this document shall be read in conjunction with other applicable ICAO DOCs and Annexes / DGCA CARs and relevant circulars, AAI CHQ/RHQ instructions on the subject issued from time to time.
4 DEFINITIONS
4.1 Aerodrome Operating Minima:
The limits of usability of an aerodrome for:
a) take off, expressed in terms of runway visual range and/or visibility and, if necessary, cloud conditions.
b) landing in 2D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude/height (MDA/H) and, if necessary, cloud conditions; and
c) landing in 3D instrument approach operations, expressed in terms of visibility and/or runway visual range and decision altitude/height (DA/H) appropriate to the type and/or category of the operation.
4.2 Instrument approach operations (Annex 6): An approach and landing using instruments for navigation guidance based on an instrument approach procedure. There are two methods for executing instrument approach operations:
a) a two-dimensional (2D) instrument approach operation, using lateral navigation guidance only; and
b) a three-dimensional (3D) instrument approach operation, using both lateral and vertical navigation guidance
Note – Lateral and vertical navigation guidance refers to the guidance provided either by:
a) a ground-based radio navigation aid; or
b) computer-generated navigation data from ground-based, space-based, self- contained navigation aids or a combination of these.
Instrument approach operations shall be classified based on the designed lowest operating minima below which an approach operation shall only be continued with the required visual reference as follows:

a) Type A: a minimum descent height or decision height at or above 75 m (250 ft);
and
b) Type B: a decision height below 75 m (250 ft). Type B instrument approach operations are categorized as:
i. Category I (CAT I): a decision height not lower than 60 m (200 ft) and with either a visibility not less than 800 m or a runway visual range not less than 550 m;
ii. Category II (CAT II): a decision height lower than 60 m (200 ft) but not lower than 30 m (100 ft) and a runway visual range not less than 300 m;
iii. Category III (CAT III): a decision height lower than 30 m (100 ft) or no decision height and a runway visual range less than 300 m or no runway visual range limitations.

4.3 Decision Altitude/Height: A specified altitude or height in a 3D instrument approach operation at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.
Note: Decision altitude (DA) is referenced to mean sea level and decision height (DH) is referenced to the threshold elevation.

4.4 ILS Critical Area: An area of defined dimensions about the localizer and glide path antennas where aircraft and vehicles are excluded during all ILS operations. The critical area is protected because the presence of vehicles / or aircraft inside its boundaries will cause unacceptable disturbance to the ILS signal-in-space.

4.5 ILS Sensitive Area: An area extending beyond the ILS critical area where the parking and/or movement of vehicles, including aircraft, are controlled to prevent the possibility of unacceptable interference to the ILS signal during ILS operations. The sensitive area is protected to provide protection against interferences cause by large moving objects outside the critical area but still normally within the airfield boundary.

4.6 Low Visibility Procedures: Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II/ III approaches and/or low visibility take-offs.

4.7 Low Visibility Take-Off (LVTO): A term used in relation to flight operations referring to a take-off on a Runway where the RVR is less than 400m.

4.8 Low-visibility operations (LVO): Approach operations in RVRs less than 550m and/or with a DH less than 60m (200 ft) or take-off operations in RVRs less than 400m.

4.9 Manoeuvring Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.

4.10 Movement Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, consisting of the manoeuvring area and the apron(s).

4.11 Runway Visual Range: The range over which the pilot of an aircraft on the centerline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centreline.

4.12 Visibility (Doc 9365): Visibility for aeronautical purposes is the greater of:
a) the greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background.
b) the greatest distance at which lights in the vicinity of 1000 candelas can be seen and identified against an unlit background.
Note 1: The two distances have different values in air of a given extinction coefficient, and the latter b) varies with the background illumination. The former a) is represented by the meteorological optical range (MOR).
Note 2: The definition applies to the observations of visibility in local routine and special reports, to the observations of prevailing and minimum visibility reported in METAR and SPECI and to the observations of ground visibility.

5 ABBREVIATIONS

5.1 The abbreviations used in descriptions of Low Visibility Procedures have the following meanings

ARFF: Aircraft Rescue & Fire Fighting
AGL: Aeronautical Ground Light
AOM: Aerodrome Operating Minima
APV: Approach with Vertical Guidance
ASMGCS: Advanced Surface Movement Guidance and Control System
ATC: Air Traffic Control
ATIS: Automatic Terminal Information Service
CFT: Crash Fire Tender
DG: Diesel Generating Set
GP: Glide Path
ILS: Instrument Landing System
IHP: Intermediate Holding Position
LOC: Localizer
LSA: Localizer Sensitive Area

LVP: Low Visibility Procedure
MID: Mid-Point
MET: Meteorology
NAV: Navigational
R/T: Radio Telephony
RVR: Runway Visual Range
SMC: Surface Movement Control
SP: Safeguarding Procedures
SSO: Shift Supervisory Officer
TDZ: Touchdown Zone
TSO: Tower Supervisory Officer
WSO: Watch Supervisory Officer

6 GENERAL

6.1 The Low Visibility Procedure (LVP) incorporates safeguarding measures to mitigate runway incursions and defines operational restrictions to ensure safe Airside Operations taking into account the available Aerodrome facilities.

6.2 Low Visibility Procedures are the procedures to ensure the safe operation of aircraft during periods of reduced visibility or low cloud base. LVP shall only be implemented when Safeguarding Procedure (SP) has been completed, and the airport is configured for low visibility operations.

6.3 An operator shall not conduct take-off with RVR/visibility less than standard Category I conditions of 550m RVR/800 m visibility unless low visibility procedures are enforced.

6.4 Category I, APV and Non-Precision Approach Operations

6.4.1 A Category I approach operation is a precision instrument approach and landing using ILS, GLS (GNSS/GBAS) or PAR with a decision height not lower than 200 ft and with visibility not less than 800 m or an RVR not less than 550 m.

6.4.2 A non-precision approach (NPA) operation is an instrument approach using any of the facilities described on Table 1 (System Minima) with a MDH or DH not lower than 250 ft and an RVR/CMV not lower than 750 m unless accepted by DGCA.

6.4.3 An APV operation is an instrument approach which utilizes lateral and vertical guidance but does not meet the requirements established for precision approach and landing operations, with a DH not lower than 250 ft and a runway visual range of not less than 600 m unless approved by DGCA.

6.4.4 Responsibility of DA/H, MDA/H, AOM, Visual Reference etc. is vested with Aircraft operator and must be approved by DGCA. Ref. DGCA All Weather CAR. Further, it is the responsibility of the service provider which publishes the approach procedure to promulgate OCA (H) for that approach procedure that forms the basis of calculation of visibility/ RVR minima by the operator.

Table 1: System Minima vs Instrument Approach Procedures (CAT I)

System Minima	
Instrument Approach Procedure	**Lowest DH / MDH**
ILS/MLS/GLS CAT I	200 ft
RNAV with approved vertical guidance	200 ft
Localizer with or without DME	250 ft
SRA (terminating at ½ NM)	250 ft
SRA (terminating at 1 NM)	300 ft
SRA (terminating at 2 NM or more)	350 ft
RNAV without approved vertical guidance	300 ft
VOR	300 ft
VOR/DME	250 ft
NDB	350 ft
NDB/DME	300 ft
VDF	350 ft

Note 1— 200 ft is the lowest authorized DH for Category I operation unless an equivalent level of safety can be achieved through use of additional procedural or operational requirements.
Note 2— A lowest DH of 200 ft for RNAV with approved vertical guidance approaches shall only be used if full SBAS capability is available. Otherwise, a DH of 250 ft is required.

6.5 Precision Approach – Category II Operations

6.5.1 A Category II operation is a precision instrument approach and landing using ILS with:
a) A decision height below 200 ft but not lower than 100 ft; and
b) A runway visual range of not less than 300 m.

6.5.2 Required RVR
The lowest minima to be used by an operator for Category II operations is 300 m for a DH of 100 ft. If it is necessary to increase DH due to, for example, facility limitations or an increased OCH, then a corresponding increase in minimum RVR will be required as shown in Table 2.

Table 2: RVR for Category II operations minima

Decision Height	Category II operations minima (RVR) coupled to below DH	
	RVR/ aeroplane Category A, B and C	RVR/ aeroplane Category D
100 ft - 120 ft	300 m	300 m / 350 m
121ft - 140 ft	400 m	400 m
141 ft – 199 ft	450 m	450 m

Note1 - The reference to "Coupled to below DH" in this table means continued use of the automatic flight control system down to a height which is not greater than 80 per cent of the applicable DH. Thus, airworthiness requirements may, through minimum engagement height for the automatic flight control system, affect the DH to be applied.
Note 2- For a CAT D aeroplane conducting an auto land, 300 m may be used.

6.6 Low visibility operations- Take Off

6.6.1 LVTO pertains to take off when the RVR is below 400 m and is applicable whenever the reported RVR in any zone (touch-down/mid/roll-out). The required facilities and conditions as mentioned in Table 3 will be as per the lowest RVR reported in any zone (e.g. if the RVR is 400/300/300 representing the three zones, then the 300 m will be the RVR for reckoning facilities and conditions of Table 3. If the RVR is 300/150/Not Reported, then 150 m will be the reckoning RVR and as the RVR is below 200 m all three RVRs are required)

6.6.2 A take-off alternate aerodrome shall be selected and specified in the operational flight plan if either the meteorological conditions at the aerodrome of departure are below the operator's established aerodrome landing minima for that operation or if it would not be possible to return to the aerodrome of departure for other reasons.

6.6.3 Take-off minima established by the operator must be expressed as visibility or RVR limits, consideration all relevant factors for each aerodrome planned to be used and the aeroplane characteristics.

6.6.4 RVR/Visibility for Take-off (Commercial Transport Aeroplanes) is as per Table 3, below.

Table 3: RVR/Visibility for Take-off (Commercial Transport Aeroplanes)

Take-off RVR/Visibility	
Facilities	**RVR/VIS Cat A, B, C & D**
Adequate Visual reference[2] (Day only)	500m
Runway edge lights or Runway Centre line markings[3]	400m
Runway edge lights and Runway Centre line markings[3]	300m

Runway edge lights and Runway Centre line lights	200m
Runway edge lights and Runway Centre line lights and relevant RVR information[4]	150m
High intensity Runway edge lights and Runway Centre line lights (spacing 15 m or less) and relevant RVR information[4]	125m
High intensity Runway edge lights and Runway Centre line lights (spacing 15 m or less), approved lateral guidance system and relevant RVR information[4]	75m
*Operation Below 200 m Subject to availability of End RVR	

Note-1: The TDZ RVR/VIS may be assessed by the pilot subject to Pilot fulfilling conditions set out in para 14,15 & 16 DGCA CAR Section 8 Series C Part –I Annexure 2
Note-1a: Pilot assessment of TDZ RVR is to be made only when the MID & ROLL OUT zone RVR are reported, and both these are not less than 200 m. (DGCA CAR Section 8 Series C Part –I Annexure 2 Para 17)
Note1b: An operator shall not conduct take-off with RVR/visibility less than standard Category I conditions of 550m RVR/ 800 m visibility unless low visibility procedures are enforced. (DGCA CAR Section 8 Series C Part –I Para 5.3)
Note1c: Below 200 m all three RVR are required. The governing RVR shall be the lowest of the reported RVRs. (DGCA CAR Section 8 Series C Part –I Table 10: RVR/Visibility for take-off)
Note-2: Adequate Visual Reference means, that a pilot is able to continuously identify the take-off surface and maintain directional control.
Note-3: For night operations at least, Runway edge lights or centre line lights and Runway end lights are available
Note-4: The required RVR must be achieved for all relevant RVR reporting points (touchdown, mid-point and stop-end/ rollout). The governing RVR shall be the lowest of the reported RVRs.
Note-5: An operator shall not conduct low visibility take-offs in less than 400m RVR unless approved by DGCA.
Note-6: An operator shall not conduct take-off with visibility / RVR less than Category-I conditions unless low visibility procedures are enforced.

7 EQUIPMENT AND SERVICES

7.1 Runway Visual Range (RVR):

a) RVR is reported in the following scales:
i) The maximum reportable value of RVR is 2000 m. When RVR is above 1500 m, it is reported as 2000 m.
ii) In the increments 100 m when greater than 800 m.
iii) In the increments 50 m when RVR greater than 400 m but less than 800 m.
iv) In the increments 25 m when less than 400 m. (as per Annex 3, Appendix 3, 4.3.6.1)
b) Equipment serviceability for CAT II operations:
i) For CAT II operations TDZ and MID RVR shall be available
ii) For LVTO, if RVR is below 200 m all three RVRs are required.

7.2 Reporting of RVR Values:

7.2.1 Various RVR equipment available at Bhopal airport to support LVP are mentioned in **Annexure 1.**
7.2.2 Touch-down zone RVR needs to be reported for CAT I operations; touch-down and mid zone RVR for CAT II operations. In all cases, touch-down zone RVR will always be controlling RVR. However, if any other RVR is reported and is relevant (operator shall not define relevant depending on Runway length/aircraft stopping distance unless approved by FSD, DGCA) it also becomes controlling. The mid zone and roll-out zone can be lower than the touch-down zone provided conditions enumerated in Note 1below are satisfied. The following Table should be used for reference:

Table 4: Reporting of RVR Values

Type of Operation	RVR		
	Touchdown Zone	Mid Zone	Roll-out zone
CAT I	550m*	125m	

CAT II	300m*	125m*	125m
*** Required for the type of operation**			

Note 1: When reporting RVR to pilots the TDZ RVR shall always be intimated for the concerned Landing RWY.
Note 2: For CAT II Operations - If TDZ RVR is below 550m then MID RVR shall also be passed.

8 MINIMUM REQUIREMENTS:

8.1 RWY 30 is equipped with ILS and is approved for CAT I and CAT II Precision Approach Lighting system. The pilot shall ensure that he is suitably qualified and certified to carry out the required category of ILS approach.

8.2 The following Aeronautical Ground lights and RVR equipment of the required standard to support Low Visibility Procedures to accommodate/permit Arrivals and/or departures in Visibility/RVR less than 800M/550M up to 350 M RVR are available at Raja Bhoj airport, Bhopal.

a) ILS localizer Transmitters and Monitors
b) ILS glide path Transmitter and Monitors
c) ILS DME
d) Standby power supply for ILS.
e) Runway edge lights,
f) Runway end lights and Threshold Identification Lights,
g) Runway Centre Line Lights,
h) CAT I & CAT II Approach Lights system, Touchdown zone lights and Side Row for runway 30,
i) Runway Stop Bar Lights & Guard Lights,
j) Instrumented RVR (touch-down & mid for CAT II).
k) UPS / Standby DG Set power for aeronautical ground lighting system.

8.3 The critical lighting system RWY edge, RWY centre line, RWY end lights and RWY 30 touchdown zone, Side Rows are supported by UPS for switch over time less than 1 second.

8.4 Real time RVR reporting System available for RWY 12-30. In case RVR equipment is unserviceable, manual RVR will be made available.

8.5 The Electrical Power Supply Systems for AGL and ILS are so designed that the time interval between failure of the primary source of power and the complete restoration of the services for associated CAT I & CAT Il shall be in accordance with maximum switch-over times in Table 8-1 of Para 8 of DGCA CAR, Section 4 Series B Part l.

8.6 In case of power failure during low visibility UPS will take over without interruption (seamless transition) without impacting any AGL facility.

8.7 When any equipment and facilities listed in 8.2 above become unserviceable during periods of LVP, the ATC unit shall advise all aircraft and accordingly suspend CAT Il Operations. Information to this effect shall be included in ATIS broadcast.

8.8 The changeover time status of various GLF facilities installed at Bhopal Airport to support LVP:

Sr.No.	Name of GLF facilities	Location	Changeover time
1.	Runway edge light fittings	Runway	Less than 1 Sec.
2.	Runway centre line fittings	Runway	Less than 1 Sec
3.	TDZ light fittings	Runway	Less than 1 Sec.
4.	Runway 30 Threshold lights	Runway	Less than 1 Sec.
5.	Runway 12 Threshold lights	Runway	Less than 1 Sec.
6.	Runway 30 END lights	Runway	Less than 1 Sec.
7.	Runway 12 END lights	Runway	Less than 1 Sec.
8.	Runway 12 approach lights (SAPL)	RWY 12 approach	Less than 1 Sec.
9.	Runway 30 approach lights (CAT-II)	RWY 30 approach	Less than 1 Sec.
10.	Runway 30 PAPI lights	Left side of RWY 30	Less than 1 Sec.
11.	Runway 12 PAPI lights	Left side of RWY 12	Less than 1 Sec.

12.	Runway 30 turn pad edge lights	RWY 30	Less than 1 Sec.
13.	Runway 12 turn pad edge lights	RWY 12	Less than 1 Sec.
14.	Taxiway edge light	Various taxiways including apron edge light	Less than 1 Sec.
15.	Signage Board	Operational area	Less than 1 Sec.
16.	Landing-T	Operational area	Less than 1 Sec.
17.	Apron Flood Lights	Old Apron & New Apron	Less than 15 Sec
18.	Isolation Bay Flood Lights	Isolation Bay	Less than 1 Sec

Note: If the intensity level of any light gets reduced to more than 50% of its minimum required level, the light shall be considered as unserviceable.

8.9 Unserviceability of Aeronautical Ground Lights/ Equipment before Implementation of LVP

8.9.1 As per Para 10.1 of DGCA CAR Section 8 Series C Part –I, A lighting system is deemed to be on outage when:
a) In the case of a lighting system comprising 6 to 13 lights (e.g. threshold lights) more than 2 lights become unserviceable, or 2 adjacent lights become unserviceable.
b) In the case of a lighting system comprising more than 13 lights, more than15% of the lights become unserviceable, or two adjacent lights become unserviceable.
8.9.2 As per Para 10.2 of DGCA CAR Section 8 Series C Part –I, The lighting system for a precision approach runway category II and/or III is deemed to be on outage when;
a) More than 5 % of the lights are unserviceable in each of the following particular significant elements;
1) precision approach category II or III lighting system, the inner 450 m;
2) runway centre line lights;
3) runway threshold lights; and
4) runway edge lights.
b) More than 10 % of the lights are unserviceable in the touchdown zone lights;
c) More than 15 % of the lights are unserviceable in the approach lighting system beyond 450 m; and
d) More than 25 % of the lights are unserviceable in the runway end lights.
e) More than two lights, or two adjacent lights of a stop bar are unserviceable.
f) Two adjacent lights of the taxiway centre line lights are unserviceable.
Note: When any two consecutive lights are unserviceable in any of the significant elements, the system is deemed to be on outage.
8.9.3 The unserviceability of the above-mentioned items will affect the operations as given below:

CRITERIA FOR CAT-II UNSERVICEABILITY

Aeronautical Ground Lighting Facility	**Un-serviceability**	**Restrictions**
Approach Lights	The inner 450 meters: more than 5 % of all Light	Suspend CAT II Operations
	Beyond 450 meters: more than 15 % of all Light	
Runway Edge lights	More than 5% of all lights	Suspend CAT II Operations
	Any two consecutive lights or more	
Runway End lights	More than 25% of all lights	Suspend CAT II Operations
	Any two consecutive lights or more	

Runway Centre Line Lights	More than 5% of all lights	Suspend CAT II Operations
	Any two consecutive lights or more	
Touch down zone lights	More than 10% of all lights	Suspend CAT II Operations
	Any two consecutive lights in Barrette	
Threshold lights	More than 5% of all lights	Suspend CAT II Operations
	Any two consecutive lights or more	
Standby Generators/UPS	Any of the generator/UPS is unserviceable	Suspend CAT II Operations

CRITERIA FOR CAT-I UNSERVICEABILITY

Aeronautical Ground Lighting Facility	Un-serviceability	Restrictions
Approach Lights	More than 15% of all lights	Suspend CAT I Operations
	Any two consecutive lights or more	
Runway Edge lights	Any two consecutive lights or more	Suspend CAT I Operations
	Any one of the two circuits serving the power supply goes unserviceable	
Runway End lights	More than 25% of all lights	Suspend CAT I Operations
	Any two consecutive lights or more	
Runway Centre Line Lights	More than 15% of all lights	Suspend CAT I Operations
	Any two consecutive lights or more	
Standby Generators/UPS	Any of the generator/UPS is unserviceable	Suspend CAT I Operations

Note 1: Under CAT Il operations, for stop bar:
a) No more than two lights will remain unserviceable; and
b) Two adjacent lights will not remain unserviceable unless the light spacing is significantly less than that specified. (Ref Para 10.5.8, DGCA- CAR Section 4 Series B Part l)
Note 2: Un-serviceability of any of the following facilities does not affect CAT II Operations:
a) PAPI
b) Taxiway clearance lights wherever provided.
Note 3: ILS sensitive and critical areas have been established for Runway 30 for CAT II & CAT-I operations. Protection of these areas during CAT-I/II operations will be ensured as per Standard Operating Procedures.
8.9.4 Preventive maintenance measure to ensure the serviceability of AGL (as applicable):
a) The system of preventive maintenance employed for a runway meant for take-off in runway visual range conditions less than a value of 550 m shall have as its objective that, during any period of operations, all runway lights are serviceable and that in any event:
i) at least 95 per cent of the lights are serviceable in the runway centre line lights and in the runway edge lights; and
ii) at least 75 per cent of the lights are serviceable in the runway end lights. In order to provide continuity of guidance, an unserviceable light shall not be permitted adjacent to another unserviceable light. (Responsibility of Electrical In-Charge).
b) The system of preventive maintenance employed for a precision approach runway category I shall have as its objective that, during any period of category I operations, all approach and runway lights are serviceable, and that in any event at least 85 per cent of the lights are serviceable in each of the following:
i) Precision approach category I lighting system;
ii) Runway threshold lights; c) Runway edge lights; and d) Runway end lights. (Responsibility of Electrical In-charge).
c) The system of preventive maintenance employed for a runway meant for take-off in runway visual range conditions of a value of 550 m or greater shall have as its objective that, during any period of operations, all runway lights are serviceable and that, in any event, at least 85 per cent of the lights are serviceable in the runway edge lights and runway end lights. In order to provide continuity of guidance, an unserviceable light shall not be permitted adjacent to another unserviceable light. (Responsibility of Electrical In-Charge).
8.9.5 In the event of failure of Runway Centre Line Light Circuit-1 during Low Visibility Operations, Runway Centre Line Light Circuit-2 shall be operated to ensure redundancy and continued availability of centre line lighting. The serviceability requirements as stipulated in Para 10.5.7 and 10.5.11 of DGCA CAR Section 4, Series B, Part I shall be ensured during CAT-II operations at the airport

9 SAFEGUARDING PROCEDURES

9.1 Safeguarding Procedures (SP) are instructions for relevant airport departments to prepare ground services and facilities for low visibility operations in order that when LVP are implemented all safeguarding procedures are complete. WSO/Duty Officer Tower will initiate and co-coordinate with all the concerned agencies for completion of safeguarding procedures before implementation of Low Visibility Procedures.
9.2 Safeguarding Procedures shall be initiated when:
a) The Visibility/RVR is less than 1200 m or visibility/RVR is forecast to deteriorate to 800 m or less; and/or
b) The cloud ceiling is less than 400 ft and forecast to fall to 200 ft or less.
9.3 Whenever meteorological conditions stipulated in para 9.2 prevails or anticipated WSO/Duty Officer Tower shall coordinate with Airport Rescue Fire Station (ARFS) for implementing the Safeguarding Procedures (SP). WSO/Duty Officer Tower shall also coordinate with following agencies for initiation of implementation/ cancellation/completion of SP as appropriate.
9.4 Coordination responsibilities are divided as follows:

9.4.1 ATC will inform:

a) Airport Rescue Fire Station (ARFS)
b) CNS (SSO/Duty Officer Equipment Room)
c) Electrical Engineering/Powerhouse
d) Duty MET Officer
e) Duty Officer Airside Operations/Management (ASM)
f) CISF Control Room

9.4.2 Airside Operations (ASM) will inform

a) Concerned Airlines
b) Terminal Manager
c) M. T. Section
d) Civil Engineering
e) IOCL & BPCL

9.4.3 The ASM will inform the above-listed agencies (Airlines, IOCL, BPCL, Civil, M.T., and T.M.) and, upon confirmation of the implementation of safeguarding procedures, will confirm it to ATC.
9.4.4 When all the concerned agencies have completed their necessary actions, they shall report to WSO/Duty Officer Tower that their Safeguarding Procedure (SP) is completed, and the airport is safeguarded for LVP operations.
9.5 Actions to be taken by Various Agencies:
9.5.1 Before commencement of winter season, a meeting will be held by Airport Director in the month of October/ November every year to inform all airlines and agencies operating at airport about their roles/ responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.
9.5.2 A refresher program on LVP for ATCO's and personnel responsible for airside operations should be conducted every year.
9.5.3 All the agencies shall ensure that staff and drivers are suitably trained for Low visibility operations.
9.5.4 WSO/Duty Officer Tower shall ensure proper log entry to be made in respect of initiation of implementation/ cancellation/completion of SP as appropriate.
9.6 Safeguarding procedures include:
a) Positioning of CFTs at the predetermined positions (i.e., at PDP-1 and PDP-2). (Responsibility of Shift In-charge Fire Station).
b) Stopping of all maintenance works on the manoeuvring area as well as removal of all men and mobile equipment from the said area. (Responsibility of Duty Officer ASM).
c) Ensuring availability of secondary power supply for change over time of maximum one second for RWY Edge, Runway Centre Line Lights and RWY End lights supported by UPS. (Responsibility of Duty Officer Powerhouse).
RWY Edge and RWY End lights may continue to operate on main power supply during safeguarding Procedures. Whenever, LVP is to be implemented as per para 9 below, the RWY Edge and RWY End lights shall be put on Standby Power Supply (DG set or UPS). This operation needs to be completed before LVP is implemented.
As UPS is available at Bhopal Airport and is capable of maintaining the required
AGL system with one second of Switch Over time with Main Supply, the main supply can continue to be primary supply, and the Generator Supply can be kept as Standby Power supply. In case of UPS is unserviceable, Generator supply will become primary source of power supply and Main power supply shall act as standby power supply.
d) Ensuring that Runways, Taxiways and other associated facilities such as Markings, Aeronautical Ground Lights are inspected within one hour preceding the implementation of LVP Operations. This is done to ensure that the lighting systems and markings are serviceable as described. Thereafter, Inspections should be done every two hours while LVP is in progress. As far as practicable, during LVP, Runway and AGL inspection will be carried out before every movement. These lighting inspections should be accorded priority and, if necessary, aircraft operations may be delayed. (Responsibility of Duty Officer ASM and Duty Officer Powerhouse).
e) Ensure all access roads are closed as mentioned in point No. f) below.
f) There are 5 Roads leading to Runway at following locations:
i) One road at Runway 30 beginning,
ii) Two separate roads at Runway 12 beginning,
iii) PDP-1, &
iv) PDP-2.
During Safeguarding Procedures, one CFT each will be positioned at PDP-1 and PDP-2. (Responsibility of Shift In-Charge Fire station).
All access roads leading to runway (other than PDP-1 and PDP-2) will remain closed with chains/barricades and Stop Lights to prevent entry of any unauthorized person or vehicle on the runway. (Responsibility of Duty Officer ASM and Duty Officer, Civil).

10 LOW VISIBILITY PROCEDURES

10.1 Implementation of Low Visibility Procedures:

10.1.1 Met officer shall inform WSO/Duty Officer Tower controller whenever Visibility/RVR reduces to 800 Meters or below and/ or cloud ceiling is at 200 ft or below.
10.1.2 Low Visibility Procedures shall be implemented when: TDZ, MID or END RVR is less than 800 m.
10.1.3 WSO/Duty officer Tower controller shall coordinate with all the agencies (as mentioned in 9.4 above) to confirm whether the Safeguarding procedures have been completed or not.
10.1.4 When Visibility/RVR falls below 800M/550M and or Cloud Ceiling is 200 ft or below and safeguarding procedures are complete, WSO/Duty Officer Tower will implement Low Visibility Procedures. WSO/Duty Officer Tower shall inform all users of the imposition of low visibility take off procedures.
10.1.5 Safeguarding Procedures (SP) and/or Low Visibility Procedures (LVP) shall be implemented at least 30 Minutes prior to the first aircraft movement or earlier based on operational requirements, as assessed and decided by the WSO/Duty Officer (ATC), when low visibility conditions are anticipated, in coordination with Duty officer (MET).
10.1.6 Subject to the condition mentioned in Para 10.1.5 above, SP/LVP may be cancelled, suspended, or terminated 15 minutes after the last arrival or 30 minutes after the last departure, depending on prevailing conditions and at the discretion of the WSO/Duty Officer (ATC) in coordination with Duty officer (MET).

(Note: Though LVP is implemented when RVR is less than 800 m, ILS CAT I operation will continue till TDZ RVR is not less than 550 m)

11 ACTIONS BY VARIOUS UNITS DURING LVP

11.1 Actions by ATC:

a) Check ILS status and Check ALCMS, whether the lighting is correctly selected and operating properly.
b) Check transmissometer displays for instrumented RVRs (MET Display).
c) Inform CNS, Aircraft Rescue & Fire Fighting Services, Airside operations, MET, CISF, Electrical, etc.
d) Make a General Broadcast on walkie-Talkie (to be monitored by concerned agencies/Personnel on Walkie-Talkie).
e) Only one aircraft shall be permitted at a time on the manoeuvring area during the time LVP is in force.
f) Towing of aircraft during LVP will be permitted only in case of extreme operational requirement. TWR shall ensure that the towing of aircraft is done under escort of "Follow Me" vehicles. "Follow Me" shall follow the route cleared by ATC.
g) TWR shall not permit any ground run on the manoeuvring area except idle power run on the stands;
h) TWR shall ensure that "Follow Me" services are provided to pilots on request;
i) The number of the vehicles on the manoeuvring area shall be restricted to bare minimum and records of all vehicles operating on the manoeuvring area shall be maintained in Delta Logbook and all vehicles operating on the manoeuvring area shall be in two-way contact with Aerodrome Control Tower.
j) The following shall be included in ATIS: "LOW VISIBILITY PROCEDURES IN FORCE".
k) Shall operate Stop Bars whenever RVR reduces to 800 meters or below.
l) Landing clearance shall be issued to the arriving aircraft when she reports established on Glidepath/ILS.
m) Landing clearance shall not be issued until:
i) Preceding landing aircraft has vacated the runway and is clear of the localizer sensitive area (LSA); or Preceding landing aircraft, carrying out missed approach, has passed over the localizer antenna.
ii) Preceding departing aircraft is airborne and has passed over the localizer antenna.
n) The ILS Critical & Sensitive Area in front of an arriving aircraft shall not be infringed from the time the arriving aircraft is expected to cross the Final Approach Point (FAP) until it has completed its landing roll.
o) TWR shall permit departures only from the beginning of the Runway in use.
p) Whenever visibility/RVR is less than 800 m/550 m, Duty officer tower shall confirm from pilot that the reported RVR value is within minima before issuing Landing or take-off clearance.
q) Duty Officer Tower shall stop all flight operations whenever RVR falls below 350 m.
r) Runway inspection should be done when interval between two aircraft movements is 30 minutes or more.
s) The bird scarers deployed in the operational area shall be specifically instructed by ATC not to come near the Runway/ Taxiway.
t) Inform changes in RVR readings to the aircraft as required.
u) Inform pilots of all failures of ILS, lighting system, Transmissiometers etc. relevant to Low Visibility Operations.
v) Initiate emergency action if aircraft on CAT I/II ILS is not in radio contact, as expected.
w) Duty officer tower shall take the Take – off Alternate from pilot before issuing take- off clearance.
x) Do not subject an aircraft to any speed control.

11.2 Actions by CNS (SSO/Duty Officer Equipment Room):

a) On receipt of "Initiating SP" from ATC, SSO/Duty Officer Equipment Room shall ensure to check ILS equipment and its status indicators in ATC units. He will inform ATC of any unserviceability in the equipment which is likely to affect ILS CAT I/II operation; & No vehicle/person is present in the sensitive/critical areas of localizer and glide path, excluding those authorized by ATC.
b) On receipt of "Initiating LVP" from ATC that LVP are to be made effective, Duty Officer will maintain continuous watch on the performance of ILS equipment and will inform ATC of any un-serviceability which may affect ILS CAT I/II operation.

11.3 Actions by Duty Manager, Airside operations (ASM):

Duty Manager shall ensure that:
a) No vehicle/person enters or is present in the sensitive/critical areas of localizer and glide path, excluding those authorized by ATC.
b) Inform concerned Airlines, T.M., M.T., Civil, IOCL & BPCL, etc.
c) All civil/electrical works in progress are to be stopped in the manoeuvring area immediately and men/material/ equipment to be removed from the sensitive/critical areas of localizer and glide path.
d) Where low visibility procedures are in effect, persons and vehicles operating on an apron shall be restricted to the essential minimum.
e) None of the workers/vehicles of grass cutting contractor, garbage removal contractor, and electrical/civil contractor shall enter the manoeuvring area during the operations of Low Visibility Procedures.
f) Duty Manager ASM shall advise Tower Supervisory Officer when all actions allocated to them are completed.
NOTE:
After ensuring above, Follow-me vehicles/operational vehicles will confirm the same to the ATC.

Subsequently, follow-me vehicle/ operational vehicles shall remain available and will maintain listening watch on Walkie-Talkie.
Any information about unserviceability of any of the Runway visual aids or power supply system will be immediately informed to ATC Tower accordingly.
No vehicles shall enter/cross in the vicinity of Runway without permission from ATC Control Tower. Such vehicles shall be able to communicate with ATC.
The vehicles of the ARFF services, civil, electrical division or of any other agency which are not equipped with RT but has to enter the Runway or taxiway for urgent operational requirement shall be escorted by the follow me jeep all the time.
Person providing "FOLLOW ME" service shall be trained and fully familiar with the taxi routes intersections and other manoeuvring area/apron/parking stands.

11.4 Action by Civil Section:

a) Civil Engg. section to ensure that all civil works in progress in manoeuvring area, are stopped and that the work area is restored in complete serviceable condition and confirm to WSO/Duty Officer ATC (Through ASM) accordingly.
b) During ILS CAT II Low Visibility Operations, no equipment, manpower or material shall be present in sensitive areas of localizer and glide path.
c) Shall ensure closure of roads leading to critical/sensitive area by chains.
d) During CAT II operations, ATC may authorize operations of vehicles Specially trained about critical and sensitive area, on manoeuvring area, including the crossing of taxiways.
e) Shall ensure that all the roads leading to runway are closed with chains/ barricades.

11.5 Action by Electrical Engg. Section/Powerhouse:

a) On receipt of advice to implement Low Visibility Procedures, Electrical Engg. Section will check that following visual aids are serviceable and can be operated at full intensity:
i) Approach lighting system
ii) Approach supplementary lighting
iii) Runway edge lights
iv) Runway Centre line lights and Stop Bar lights
v) Runway threshold and Runway end lights

NOTE 1: *The Electrical Department Personnel shall be present in Control Tower during periods of reduced visibility and ensure that the ALCMS display(s) in the Control Tower are serviceable.*
NOTE 2: *No adjustment in light intensity shall be made without permission from ATC Tower during LVP.*
b) They shall ensure that no electrical maintenance works is carried out during LVP either in powerhouse or on any other electrical facilities used during CAT II operations.
c) Ensure that Runways, Taxiways and other associated AGL facilities are inspected every 2 hours while LVP is in progress.
d) As far as practicable, during LVP, Runway and AGL inspection will be carried out before every movement.
e) They will inform the un-serviceability or any change in status of any facility/systems to ATC & Duty Manager Airside Ops immediately.

11.6 Actions by ARFF:

a) Positioning of crash fire tenders on pre-determined positions (i.e., at PDP-1 and PDP- 2).
b) ARFF/ CFTs must obtain clearances from ATC prior to entry to any aircraft movement area other than the designated service roads.

11.7 Actions by CISF:

a) The officer in charge on duty shall ensure by deploying adequate manpower that vital electrical installations are properly secured and protected against any unauthorized intrusion.
b) The Officer in charge, on receipt of advice to implement Low Visibility Procedures, will immediately inform all access gates and CISF posts under their respective controls in operational area to restrict movement of vehicles under their control.

11.8 Actions by Duty Officer (Meteorological Office):

a) Duty Meteorological Officer would issue an ‘Outlook for Low Visibility Procedures’ to the ATC whenever he expects that the RVR and/or cloud ceiling will fall below 800 m and/or 400 ft or less respectively.
b) Whenever Duty Meteorological Officer visualizes that RVR is likely to fall below 800 m and/or cloud ceiling to 200 ft or less within next 2 hours, he will issue an 'Advisory Message' to ATC to this effect.
c) When the RVR and/or cloud ceiling are 800m and/or 200ft respectively and the trend is towards improvement in these elements of weather conditions the Duty Met Officer may, when requested by ATC, advise him about such improving weather conditions for the purpose of termination of LVP.
d) The Duty Met Officer shall be present in Control Tower during periods of reduced visibility and ensure that the RVR displays in the Control Tower is serviceable.
e) He would also ensure that RVR/visibility recorders of Touch-down zone and Midpoint are serviceable.
f) In case, Instrumented RVR is not available, real time RVR shall be provided, whenever requested. For this purpose, Air Side Operation shall provide necessary vehicle whenever required and inform CISF to give priority to the Met team to avoid any delay because of security checks near entry gate. This may also include requirement of female CISF frisking staff to facilitate quick entry of female IMD staff through Airside entry Gate.

11.9 Actions By Other Agencies (Airlines, Refuelling Companies, Catering Agencies, Etc.)

a) Every year before commencement of monsoon/winter season, a meeting will be held by Airport Director, to inform all airlines and agencies operating at airport about their roles/responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.

b) All the agencies shall ensure that staff and drivers are suitably trained during Low Visibility Procedure operations.

c) All agencies operating in the operational area shall ensure that only those vehicles that are absolutely essential for aircraft operations operate in the operational area during periods of low visibility. The drivers of these vehicles should keep a look out for taxiing aircraft and other vehicles to prevent accidents.

d) All the vehicles must have their obstruction lights “ON” during Low Visibility Procedures operations and strictly follow vehicular lane.

e) All instructions/sign boards provided for vehicular movement area/service roads must be followed while operating in the operational area.

f) No vehicle/equipment/personnel shall enter in and around the vicinity of the runways or taxi-tracks except with prior permission of Duty Officer (TOWER).

g) Airlines/ground handling agencies must ensure that push back area and taxiing path of aircraft on apron shall be kept clear of equipment/vehicles.

12 TERMINATION OF LOW VISIBILITY PROCEDURES

a) When facilities and equipment necessary for CAT I/CAT Il operations for RWY 30 are degraded and/or the prevailing conditions are considered unsafe for such operations.

b) When Visibility/RVR improves to 800M/550 M or more and cloud ceiling is 200 feet or higher and trend is for improvement, WSO/Duty Officer Tower would terminate operations of LVP. He may obtain advice from Duty MET Officer regarding improvement in weather conditions before the termination of LVP.

c) The WSO/Duty Officer Tower will intimate the following regarding termination of operations:

i) ARFF
ii) Duty Met Officer
iii) CNS (SSO/Duty Officer Equipment Room)
iv) Airside Operations
v) CISF
vi) Electrical Engg.

d) ASM will intimate all previously notified personnel (undermentioned) to resume normal operations:

i) Terminal Manager
ii) Civil Engg.
iii) Airlines
iv) IOCL & BPCL

e) On cancelling of LVP, following message shall be included in two subsequent ATIS broadcasts. “LOW VISIBILITY PROCEDURES CANCELLED”.

f) If SP are implemented and LVP are not subsequently implemented and the visibility/RVR improves and is more than 1200 m and/or the cloud ceiling is 400ft or higher and both are forecast to remain above the required SP criteria, WSO/Duty Officer Tower may cancel SP.

VABP AD 2.23 ADDITIONAL INFORMATION

1. Declared Distance for departure RWY 30 from TWY E Intersection:

   TORA / TODA: 2239M

2. Runway Turn Pads, capable of handling B777-300 ER type of aircraft, provided at both ends of Runway for Special Operations with prior permission.

VABP AD 2.24 CHARTS RELATED TO AN AERODROME

1. Aerodrome Chart (01 OCT 2026)
2. Aircraft Parking/Docking Chart (02OCT 2025)
3. Aerodrome Obstacle Chart Type – A (Operating Limitations) RWY 12 (02OCT 2025)
4. Aerodrome Obstacle Chart Type – A (Operating Limitations) RWY 30 (02OCT 2025)
5. Aerodrome Obstacle Chart Type - B (08 August 2024)
6. VOR Procedure RWY 12 (08 August 2024)
7. VOR Procedure RWY 30 (08 August 2024)
8. RNP Y Procedure RWY 12 (08 August 2024)
9. Coding Table - RNP Y Procedure RWY 12 (08 August 2024)
10. RNP Y Procedure RWY 30 (08 August 2024)
11. Coding Table - RNP Y Procedure RWY 30 (08 August 2024)
12. ILS or LOC Procedure RWY 30 (17 April 2025)
13. ILS Critical And Sensitive Area (19 March 2026)

VABP AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

Procedure	Procedure Minima	VSS Penetration
1	**2**	**3**